

B. Truth Discovery Based on Cryptography

Substantial research has been conducted on privacy-preserving truth discovery using cryptographic techniques. Zhang et al. [11] developed two methods using Paillier homomorphic encryption for both stable and dynamic worker locations. Xue et al. [41] combined homomorphic encryption with two non-colluding servers and a weight-based incentive mechanism. Zhao et al. [42] contributed a robust anonymization mechanism within an edge computing paradigm. Wu et al. [39] proposed both a basic privacy-aware approach and an advanced privacy-enhanced version with two servers. Recent advances have explored alternative strategies for enhancing privacy in truth discovery. Tang et al. [40] developed an anonymized method that decouples workers from their data, alongside a lightweight perturbation method. Zhang et al. [43] designed a method using matrix computation and key derivation mechanisms. Liu et al. [44] introduced a collaborative cloud-encrypted system with garbled circuits, while Xiong et al. [45] proposed a decentralized framework combining secret sharing with additive homomorphic encryption. Liu et al. [46] further advanced these approaches by integrating data masking techniques for spatiotemporal contexts. Liang et al. [47] proposed a privacy-preserving and reliable truth discovery framework tailored for heterogeneous fog-based crowdsensing environments, addressing data variability and trustworthiness in decentralized settings. Bai et al. [48] introduce ETBP-TD, an efficient bilateral privacy-preserving scheme for mobile crowdsensing, ensuring both data contributor and aggregator privacy while maintaining high truth discovery accuracy.

Nevertheless, these cryptographic approaches typically rely on trusted third parties, which may not be practical in real-world deployments. They also incur substantial computational and communication overhead due to complex encryption operations. Since we resort to LDP for protecting *identity privacy* without cryptographic techniques, these valuable contributions are orthogonal to OSCAR taken here.

C. Truth Discovery Under LDP

The integration of LDP into truth discovery has evolved along two primary research trajectories, based on the nature of input data: discrete and continuous inputs. This categorization reflects the distinct technical challenges and privacy preservation approaches required for each data type.

Discrete Inputs: For discrete data scenarios, researchers have developed several foundational approaches. Li et al. [7] pioneered a two-layer framework that achieves enhanced utility across most use cases. Building on this work, Sun et al. [2] introduced personalized incentive mechanisms specifically designed for binary-choice questions. Wang et al. [8] advanced the field by developing a method based on local attribute differential privacy, though their approach relies on the assumption that privacy sensitivity is limited to a subset of attributes.

Continuous Inputs: The continuous data domain has seen more diverse developments, reflecting its greater complexity. Li et al. [9] extended their earlier work by incorporating Gaussian mechanisms to achieve (ϵ, δ) -LDP, albeit with probabilistic guarantees that represent a relaxation of strict LDP requirements.

TABLE II
SUMMARY AND COMPARISON

Approaches	Rigorous LDP	Data Privacy	Distance Measurement	Identity Privacy
[7], [2]	✓	✓		
[8]		✓		
[9], [10]		✓		
[4]				✓
[11]	✓	✓		
[12]	✓	✓	✓	
[13], [14]		✓		
[15]	✓	✓		
OSCAR	✓	✓	✓	✓

Sun et al. [4] introduced matrix factorization-based perturbation, achieving ϵ -cell LDP, though this also represents a weaker form of privacy protection. Further advances came from Sun et al. [10], who addressed incentive challenges through a personalized approach providing ϵ -KL-LDP guarantees. Notable progress in maintaining rigorous LDP while improving accuracy was achieved by Zhang et al. [11], [12] through innovative sampling and probabilistic estimation techniques.

The field has also expanded to address streaming data scenarios. Huang et al. [13] developed methods for handling continuous data streams using sliding windows and Gaussian mechanisms, achieving w -event- (ϵ, δ) -LDP. Wang et al. [14] contributed specialized approaches for edge computing environments under w -event- (ϵ, δ) -DP constraints. Pang et al. [15] combined personalized incentives with privacy preservation to enhance accuracy while maintaining ϵ -DP guarantees.

We summarize them and compare them with our OSCAR solution in Table II. There are four key criteria in this comparison: Rigorous LDP indicates whether an approach provides strict ϵ -local differential privacy guarantees without relaxation, ensuring the strongest level of privacy protection for individual workers. Data Privacy refers to the protection of the actual values or responses submitted by workers for each task. Distance Measurement denotes whether the approach considers the statistical characteristics of injected noise when measuring differences between reported values and ground truths. Identity Privacy indicates whether the method protects information about which specific tasks a worker chooses to participate in, preserving privacy around their task selection patterns.

The systematic comparison in Table II reveals several critical gaps in existing truth discovery approaches under LDP. We can examine these limitations across four key dimensions of privacy protection and utility. The first major gap concerns the level of privacy guarantee. Many existing solutions [4], [8], [9], [10], [13], [14] do not provide rigorous LDP protection. This represents a significant challenge in balancing strong privacy guarantees with practical utility. The second limitation stems from the lack of comprehensive privacy protection in existing work. Most approaches focus narrowly on either data privacy or identity privacy rather than addressing both concerns. For example, the methods proposed in [2], [7] and [11] address data privacy but neglect identity privacy, while [4] considers identity